A logic of knowledge based on abstract arguments

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Abstract

We introduce a logic of knowledge in a framework in which knowledge is treated as a kind of belief. The framework is based on a standard KD45 characterization of belief, and the characterization of knowledge undergoes the classical tripartite analysis that knowledge is justified true belief, which has a natural link to the studies of logics of evidence and justification. The interpretation of knowledge avoids the unwanted properties of logical omniscience, independent of the choice of the base logic of belief. We axiomatize the logic, prove its soundness and completeness and study the computational complexity results of the model checking and satisfiability problems. We extend the logic to a multi-agent setting and introduce a variant in which belief is characterized in a weaker system to avoid the problem of logical omniscience.

Keywords: Knowledge, belief, epistemic logic, doxastic logic, justified true belief, computational complexity

1 Introduction

Modern studies in epistemic logic and doxastic logic were initiated in [31, 60] and later examined extensively in [23, 40]. Most variants of epistemic and doxastic logics do not mix the two notions. Although there are traditions characterizing both knowledge and belief in one framework (say, [34, 50]), this has received far less attention compared with the efforts made in either side.

For the literature on the logics of both knowledge and belief, and in particular of the relationship between them, there are various approaches that can largely be classified into three categories: (i) that treating both knowledge and belief as primitive concepts, (ii) where knowledge or belief is a derived concept from the other and (iii) where both knowledge and belief are derived concepts. The famous tripartite definition of knowledge dating back to Plato, that knowledge is 'justified true belief' (also known as the JTB paradigm), falls into the second category (see, e.g. [37] for an implementation in logic). Our work lies in this category as well. We shall give a closer look into the literature in Section 6.

We introduce a logic LK based on a classical modal logic of belief, in which a KD45 relation (a binary relation that is serial, transitive and Euclidean) is used for characterizing belief.[1] We interpret knowledge as 'true belief that has an appropriate argument', where an *argument* for it is characterized

[1] The base logic KD45 for characterizing belief is not fundamental here. An alternative base logic will be introduced in Section 5.2.

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by a collection of possible situations (sets of worlds) with one of which representing the true belief, and the argument is *appropriate* if all the basic facts that is represented by a situation are beliefs. This intuitively relates to the work on logical approaches in the justification of knowledge [11, 52], in particular, the fruitful direction of justification logic [3, 4, 24], and we will discuss their differences in Section 6.

The application of justification logic in the discipline of epistemic logic [6, 7] connects to the problem of logical omniscience. In normal modal systems, all logical validities are *necessarily* true, since the technical characterization in a normal epistemic logic makes it true that the modal box operator is closed under logical consequence. This becomes a problem when the modal box is used to express knowing and believing in epistemic and doxastic logics. Classical modal systems, such as those between S4 and S5, which are typically used to characterize knowledge and those like KD45 and K45 for belief, all have the problem of logical omniscience. This problem has been recognized already in [31, page 31], and we refer to [48, 49] for more details. One of the main solutions to this problem was to view knowledge or belief defined in this way to be *implicit* or *potential* (or more generally, some concept that is not knowledge or belief, but closely related), that one can only obtain in an ideal case, such as having an extreme power of reasoning. Only when the one is *aware* of the implicit knowledge or belief, it becomes *explicit*. This tradition has attracted a lot of attention since the seminal work on the logic of (un)awareness [22].

There have been solutions to the problem of logical omniscience without using the notion of awareness, since very early [19, 32] to very new attempts [12]. For example, [59] introduces a logic of knowledge that does not have the problem by means of requiring the truth of p and the truth of its 'epistemic counterparts' in the interpretation of 'knowing p'. With a certain similarity that is to be discussed in Section 6, our framework arguably avoids the problem of logical omniscience in the interpretation of knowledge.[2]

By proposing the logic LK we intend to provide a conceptually eligible, simple and flexible logic framework for representing knowledge: *eligible* as the interpretation of knowledge in line with the JTB paradigm[3] ; *simple* not only in the intuitive understanding, but also technically, as we will show that the model checking problem for LK is in P and the satisfiability problem for LK is NP-complete; *flexible* for the interpretation of knowledge being independent from that of belief, which will be discussed in Section 5.2.

The paper is presented as follows. We give definitions, explanations and basic results of the logic LK in the next section. An axiomatization of LK is introduced in Section 3, and we show its soundness and completeness there. In Section 4, we study the complexity of the model checking and satisfiability problems for LK. We show that the model checking problem is in P and the satisfiability problem is NP-complete. In Section 5 we extend LK to characterize multiple agents and give an alternative approach whose belief operator does not suffer from the logical omniscience problem. We discuss related work in Section 6 and conclude in Section 7.

[2]The purely semantic interpretation of arguments, as we use, cannot discern between syntactically different formulas that describe the same situation—a touchstone of this is the validity of the rule of replacement of equivalents—which is sometimes viewed as an example of the problem of logical omniscience; we will address this issue in Section 6 where we compare our work with those in the direction of justification logic.

[3]Though the JTB paradigm has been criticized by the epistemologists for being insufficient [26], it is a more sophisticated definition of knowledge for epistemic logicians.

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2 The logic LK

In this section we introduce a logic LK for reasoning about knowledge as a type of true belief with an appropriate argument, based a classical modal logic of belief. We first introduce the language of LK, by assuming a countably infinite set P of propositional variables.

DEFINITION 2.1
(languages).
The language $\mathcal{L}$ for LK is given by the following grammar rule:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \wedge \varphi) \mid B\varphi \mid K\varphi,$$

where $p \in P$. Moreover, let $\mathcal{PL}$ be the language of propositional logic, i.e. the sublanguage of $\mathcal{L}$ without the operators B and K.

$B\varphi$ and $K\varphi$ are read as 'the agent *believes* φ' and 'the agent *knows* φ', respectively. For simplicity, here we focus on a single-agent system. We shall study a multi-agent extension in Section 5.1. Other propositional connectives are defined by usual abbreviations.

Now we introduce the formal models for LK.

DEFINITION 2.2
(models).
A *model* is a quadruple $M = (W, R, S, V)$ such that:

- W is a non-empty set of (possible) worlds (or states),
- $R \subseteq W \times W$ is a binary relation that is serial, transitive and Euclidean,
- $S : W \to \wp(\wp(\wp(W)))$ is an *argumentation function*, and
- $V : P \to \wp(W)$ is a valuation that assigns every propositional variable a set of possible worlds.

As usual, M, together with a state w of it, forms a *pointed model* (M, w).

In the above definition, it is clear that (W, R, V) forms a standard Kripke model, more precisely, a *KD45 model*, as the relation R is serial, transitive and Euclidean. This follows the classical way of modeling belief [23, 40]. What deserves extra explanation is the argumentation function S, which maps a state to a set of sets of sets of states. This has a flavor of neighbourhood semantics [18].[4] Intuitively, a set of states will be used to stand for the states where a formula is true. A set of formulas is understood as a argument (or proof). Hence S will be used in an interpretation to link a state to a set of arguments, which will play an important role in the interpretation of knowledge.

Let us first introduce the formal semantics.

[4]In the neighbourhood semantics, a neighbourhood function maps a state to a set of *neighbourhoods* (sets of states), and these neighbourhoods are used to model the statements that are *necessary* (or *known*, in the context of interpreting knowledge) in the state. An argumentation function here maps a state to a collection of sets of 'neighbourhoods' that play a similar role, except that the argumentation function is of higher order because an argument is in some sense a collection of statements.

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DEFINITION 2.3
(satisfaction).
Given a model $M = (W, R, S, V)$ and a world $w \in W$, that a formula α is *satisfied* (or *true*) in the pointed model (M, w) (notation: $M, w \models \alpha$) is inductively defined as follows:

$$
\begin{array}{lll}
M, w \models p & \text{iff} & w \in V(p) \\
M, w \models \neg\varphi & \text{iff} & \text{not } M, w \models \varphi \\
M, w \models (\varphi \wedge \psi) & \text{iff} & M, w \models \varphi \text{and } M, w \models \psi \\
M, w \models B\varphi & \text{iff} & \text{for all } u \in W, \text{ if } wRu \text{then } M, u \models \varphi \\
M, w \models K\varphi & \text{iff} & M, w \models \varphi \wedge B\varphi \text{ and there exists } X \in S(w) \text{such that:} \\
 & & [\![\varphi]\!]_M \in X \text{ and for all } \psi \in \mathcal{PL} \text{ with } [\![\psi]\!]_M \in X, M, w \models B\psi
\end{array}
$$

where $[\![\varphi]\!]_M = \{x \in W \mid M, x \models \varphi\}$ denotes the *truth set* of φ in M (similarly for the truth set of ψ). We omit the subscript M when it is clear from the context.

A formula φ is called *valid*, denoted $\models \varphi$, if for all pointed models (M, w), $M, w \models \varphi$; otherwise it is called *invalid*. For any set Γ of formulas, we say φ is a semantic consequence of Γ, denoted $\Gamma \models \varphi$, if all pointed models (M, w) that satisfies all formulas of Γ also satisfy φ.

The interpretation of $K\varphi$ follows the tripartite definition of knowledge, that knowledge is 'justified true belief', as it can easily be observed from the interpretation that the truth of $K\varphi$ implies the truth of φ and $B\varphi$, and that there is an argument for φ. More precisely, $K\varphi$ is true, if φ is true and believed to be so, and there is an *appropriate* argument for it, in the sense that φ is consistent with the argument and all propositional formulas of that argument must be believed to be true.

A technical issue here is that we could not enforce the belief of all formulas in an argument, or otherwise the inductive definition contains a vicious circle. A typical solution is to consider only simpler formulas (say, according to modal depths), but to make it simple in this paper we consider only propositional formulas and the relevant formula φ. This does get an intuitive explanation: beliefs and disbeliefs are not so important for an argument compared to basic facts.

EXAMPLE 2.4
Consider an agent who *believes* that the number 47 is a prime and all her argument for this is that 47 is not divisible by 7. We may all agree that the agent does not *know* that 47 is a prime since the composite 8 is also not divisible by 7 and the agent's belief happens to be true.

Let p denote the proposition '47 is a prime' and q '8 is a prime', respectively. In the above setting, we see that the agent proposes an argument for p, i.e. every number not divisible by 7 is a prime. In addition to p, this argument has the other consequence q, which means that the 'X' for representing this argument comprises the truth sets of p and q (Of course there are other consequences following from this argument, e.g. 1 is a prime, 2 is a prime and so on. But for simplicity, we have limited our language here). So the whole scenario can be modeled in our framework by a model $M = (W, R, S, V)$ with $W = \{w\}$, $R = \{(w, w)\}$, $S(w) = \{\{\{w\}, \{\}\}\}$, $V(p) = \{w\}$ and $V(q) = \emptyset$. The *frame* (W, R) on which M is based is typically represented by the following diagram:

It is not hard to verify that $M, w \models Bp$. Now one can see that $M, w \not\models Kp$, because the only 'argument' containing $[\![p]\!]_M = \{w\}$ is $\{\{w\}, \{\}\}$, but it is not appropriate (since $[\![q]\!]_M = \{\} \in \{\{w\}, \{\}\}$ and $M, w \not\models Bq$).

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We list some (in)validities of the logic regarding the characterization of knowledge, including those for characterizing that knowledge is true belief.

PROPOSITION 2.5
(LK knowledge).
The following properties hold for all formulas φ and ψ:

1. (verity) $\models K\varphi \to \varphi$
2. (being belief) $\models K\varphi \to B\varphi$
3. (replacement of equivalents) if $\models \varphi \leftrightarrow \psi$ then $\models K\varphi \leftrightarrow K\psi$

The following hold for some formulas φ and ψ:

4. (lack of distribution over implication) $\not\models K(\varphi \to \psi) \to (K\varphi \to K\psi)$
5. (lack of positive introspection) $\not\models K\varphi \to KK\varphi$
6. (lack of negative introspection) $\not\models \neg K\varphi \to K\neg K\varphi$
7. (lack of generalization) Even if $\models \varphi$, it is not necessarily $\models K\varphi$
8. Even if $\models \varphi \to \psi$, it is not necessarily $\models K\varphi \to K\psi$.

PROOF. The first two clauses can be seen easily from the definitions.

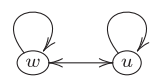
3. Assume $\models \varphi \leftrightarrow \psi$. Let $M = (W, R, S, V)$ be a model and $w \in W$. We need to show that $M, w \models K\varphi \leftrightarrow K\psi$, namely $M, w \models K\varphi$ if and only if $M, w \models K\psi$. Without loss of generality, suppose $M, w \models K\varphi$, and it suffices to show that $M, w \models K\psi$. By definition $M, w \models \varphi \wedge B\varphi$ and there is an appropriate argument X for φ, in the sense that $X \in S(w)$ and $[\![\varphi]\!] \in X$ and $\forall \chi \in \mathcal{PL}$: $[\![\chi]\!] \in X \Rightarrow M, w \models B\chi$. By the assumption we have $[\![\varphi]\!] = [\![\psi]\!]$. Moreover, $\models B\varphi \leftrightarrow B\psi$ (B is a standard KD45 modal operator). Therefore, $M, w \models \psi \wedge B\psi$ and X is also appropriate for ψ. This shows $M, w \models K\psi$, and thus $M, w \models K\varphi \leftrightarrow K\psi$.

4. We give a countermodel. Consider the model $M_1 = (W_1, R_1, S_1, V_1)$ such that $W_1 = \{w, u, v\}$, $R_1 = \{(w, w), (u, u), (v, v)\}$, with (W_1, R_1) illustrated below:

and $V_1(p) = \{w\}$, $V_1(q) = \{w, v\}$, $S_1(w) = \{\{\{w, u, v\}\}, \{\{w\}\}\}$ and $S_1(u) = S_1(v) = \emptyset$.

We have $M_1, w \models K(p \to q)$, because (1) $M_1, w \models (p \to q) \wedge B(p \to q)$ and (2) $\{\{w, u, v\}\}$ is an appropriate argument since $[\![p \to q]\!]_{M_1} = \{w, u, v\} \in \{\{w, u, v\}\}$ and $M_1, w \models B\psi$ for any $\psi \in \mathcal{PL}$ with $[\![\psi]\!]_{M_1} \in \{\{w, u, v\}\}$ (for $\{x \in W_1 \mid wRx\} \subseteq \{w, u, v\}$). Similarly, we have $M_1, w \models Kp$, but $M_1, w \not\models Kq$. So, $M_1, w \not\models K(p \to q) \to (Kp \to Kq)$.

5. Consider a countermodel $M_2 = \{W_2, R_2, S_2, V_2\}$ with (W_2, R_2) illustrated below:



$S(w) = \{\{\{w, u\}\}\}$, $S(u) = \emptyset$, and $V(p) = \{w, u\}$. We have $M_2, w \models Kp$ and $M_2, u \not\models Kp$. Thus $M_2, w \not\models BKp$. Since $\models KKp \to BKp$, $M_2, w \not\models KKp$.

6. Consider a countermodel $M_3 = \{W_3, R_3, S_3, V_3\}$ with (W_3, R_3) illustrated below:

$S_3(w) = \emptyset$ and $V_3(p) = \emptyset$. We have $M_3, w \models \neg Kp$ but $M_3, w \not\models K\neg Kp$.

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Axioms:	
PC	All the instances of propositional tautologies
K	$B(\varphi \to \psi) \to (B\varphi \to B\psi)$
D	$B\varphi \to \neg B\neg\varphi$
4	$B\varphi \to BB\varphi$
5	$\neg B\varphi \to B\neg B\varphi$
KB	$K\varphi \to B\varphi$
TK	$K\varphi \to \varphi$
Rules:	
N	from φ infer $B\varphi$
RE	from $(\varphi \leftrightarrow \psi)$ infer $(K\varphi \leftrightarrow K\psi)$
MP	from φ and $(\varphi \to \psi)$, infer ψ

FIGURE 1 The axiomatization **LK**

7. The model M_3 introduced in clause 6 is a suitable countermodel, since $\models \neg(p \wedge \neg p)$ and $M_3, w \not\models K\neg(p \wedge \neg p)$.

8. It is clear that $\models p \to (p \vee q)$. Using the model M_1 in clause 4, we can show that $M_1, w \models Kp$. We can also show that $M_1, w \not\models K(p \vee q)$. Therefore $\not\models Kp \to K(p \vee q)$. □

REMARK 2.6
The lack of generalization and distribution over implication for knowledge is what we would like to see, for these are direct causes of the problem of logical omniscience. The reader may instead want to have the principles of positive and negative introspection. Here we presented a logic without these principles mainly for the purpose of pursuing a simple and flexible framework. If one would like to enforce, e.g. positive introspection, it can be done by adding the following constraint on the argumentation function S in a model: for all worlds w, for all $X \in S(w)$,

1. for all formulas φ, if $[\![\varphi]\!] \in X$ then $[\![K\varphi]\!] \in X$, and
2. for all worlds u such that wRu, if $X \in S(w)$, then $X \in S(u)$.

Moreover, the weak logic LK does not enforce some relationships between knowledge and belief either. For example, positive and negative introspection ($B\varphi \to KB\varphi$, $\neg B\varphi \to K\neg B\varphi$, $K\varphi \to BK\varphi$ and $\neg K\varphi \to B\neg K\varphi$) are not valid.

3 Axiomatization

In this section we introduce a sound and complete axiomatization of the logic LK, i.e. **LK**, which is given in Figure 1.

The axioms PC, K, D, 4 and 5 and the rules N and MP form an axiomatization of the standard KD45 logic of belief. There are a few more axioms and rules for the characterization of knowledge.

The soundness of **LK** is not hard to verify. All the axioms are valid, and the rules preserve validity, some of which are shown in Proposition 2.5. As for the completeness, we prove by the canonical model method, and this will be our main task for the rest of this section.

We refer to a modal logic textbook (say, [14]) for the definitions of *proof*, *deduction* and a *(maximal) consistent* set of formulas in an axiomatization. A well-known Lindenbaum's Lemma

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claims that any consistent set of formulas is a subset of a maximal consistent set of formulas, which also holds for **LK**.

As a notational convention, the set of all maximal consistent sets of $\mathcal{L}$ formulas is denoted MCS. Moreover, we shall write $[\![\varphi]\!]^c$ for the set of all maximal consistent sets of formulas containing φ, i.e. $[\![\varphi]\!]^c = \{\Gamma \in MCS \mid \varphi \in \Gamma\}$.

We first introduce some properties of maximal consistent sets of formulas.

LEMMA 3.1
Let $\Gamma \in MCS$. The following hold:

1. Γ is closed under modus ponens, i.e. if $\varphi, (\varphi \to \psi) \in \Gamma$, then $\psi \in \Gamma$;
2. For all formulas φ, either $\varphi \in \Gamma$ or $\neg\varphi \in \Gamma$, but not both;
3. $[\![\varphi]\!]^c = [\![\psi]\!]^c$ if and only if $\varphi \leftrightarrow \psi$ is an **LK** theorem.

PROOF. The first two clauses can be shown in the same way as in the literature. Now for the third:

From left to right. Suppose $\nvdash_{\mathbf{LK}} \varphi \leftrightarrow \psi$, then $\{\varphi, \neg\psi\}$ or $\{\neg\varphi, \psi\}$ is consistent. Without loss of generality, we consider the first case, i.e. $\{\varphi, \neg\psi\}$ is consistent. Then there is $\Delta \in MCS$ such that $\{\varphi, \neg\psi\} \subseteq \Delta$. By the second clause of this lemma, $\psi \notin \Delta$, which implies that $[\![\varphi]\!]^c \neq [\![\psi]\!]^c$, as is to be shown.

From right to left. Suppose $[\![\varphi]\!]^c \neq [\![\psi]\!]^c$ and $\vdash_{\mathbf{LK}} \varphi \leftrightarrow \psi$. Without loss of generality, consider the case when there exists $\Delta \in MCS$ such that $\varphi \in \Delta$ and $\psi \notin \Delta$. It follows that $\nvdash_{\mathbf{LK}} \varphi \leftrightarrow \psi$, for otherwise there is a violation of the first and second clauses of this lemma. □

DEFINITION 3.2
(canonical model).
The *canonical model* for **LK** is the structure $M^c = (W^c, R^c, S^c, V^c)$ where:

- $W^c = MCS$, i.e. the set of all maximal consistent sets of formulas;
- R^c is binary relation on W^c defined by: $(w, u) \in R^c$ iff for all formula φ, if $B\varphi \in w$ then $\varphi \in u$;
- $S^c : W^c \to \wp(\wp(\wp(W^c)))$ is such that $S^c(w) = \{\{[\![\varphi]\!]^c\} \mid K\varphi \in w\}$;
- V^c is the valuation defined by: $V^c(p) = [\![p]\!]^c$ for all $p \in P$.

It is easy to see that $M^c = (W^c, R^c, V^c)$ forms the canonical model for the standard KD45 logic. Since **LK** includes an axiomatization of KD45 as a subsystem, we can use the classical method to show that R^c is serial, transitive and Euclidean (cf., say, [14]). Moreover, S^c matches the type of an argumentation function. It is not hard to verify that the canonical model is a model of LK.

LEMMA 3.3
(truth).
For any formula $\varphi \in \mathcal{L}$ and any $w \in MCS$:

$$\mathcal{M}^c, w \models \varphi \quad \text{iff} \quad \varphi \in w.$$

PROOF. First of all, we can define the modal degree of any formula $\varphi \in \mathcal{L}$, notation $d(\varphi)$, as follows:

$$\begin{array}{rcl} d(p) & = & 0 \\ d(\neg\varphi) & = & d(\varphi) \\ d((\varphi \wedge \psi)) & = & \max(d(\varphi), d(\psi)) \\ d(B\varphi) & = & d(\varphi) + 1 \\ d(K\varphi) & = & d(\varphi) + 1. \end{array}$$

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Our proof is carried out by nested induction on $d(\varphi)$ and within that on the structure of φ. The case $d(\varphi) = 0$ is trivial. Suppose for every φ such that $d(\varphi) \leq n$: $\mathcal{M}^c, w \models \varphi \Leftrightarrow \varphi \in w$. We are going to show that the same holds for all φ such that $d(\varphi) \leq n + 1$. The only interesting case is the inductive step for $K\varphi$:

From left to right. Suppose $K\varphi \notin w$. We claim that $\{[\![\varphi]\!]^c\} \notin S^c(w)$. Suppose not, there must be a ψ such that $K\psi \in w$ and $[\![\psi]\!]^c = [\![\varphi]\!]^c$. This implies that $\vdash_{\mathbf{LK}} \varphi \leftrightarrow \psi$ by Lemma 3.1(3). Therefore $\vdash_{\mathbf{LK}} K\varphi \leftrightarrow K\psi$ by rule the RE. But $K\varphi \notin w$, this contradicts the fact that w is closed under modus ponens. Thus $\{[\![\varphi]\!]^c\} \notin S^c(w)$. Since $d(\varphi) < d(K\varphi)$, by the induction hypothesis, $[\![\varphi]\!] = [\![\varphi]\!]^c$, therefore $\{[\![\varphi]\!]\} \notin S^c(w)$, and so $\mathcal{M}^c, w \not\models K\varphi$.

From right to left. We need to show that (1) $\mathcal{M}^c, w \models \varphi \wedge B\varphi$ and (2) there is X in $S^c(w)$ such that $[\![\varphi]\!] \in X$ and $\forall \psi \in \mathcal{PL} : [\![\psi]\!] \in X \Rightarrow \mathcal{M}^c, w \models B\psi$. For (1), from $\vdash_{\mathbf{LK}} K\varphi \rightarrow \varphi$, it follows that $\varphi \in w$. Since $d(\varphi) < d(K\varphi)$, $\mathcal{M}^c, w \models \varphi$ by the induction hypothesis. On the other hand, since $\vdash_{\mathbf{LK}} K\varphi \rightarrow B\varphi$, we have $B\varphi \in w$. We can show that $M^c, w \models B\varphi$ in the same way as for a normal modal logic. For (2), since $K\varphi \in w$, $\{[\![\varphi]\!]^c\} \in S^c(w)$. We claim that $\{[\![\varphi]\!]^c\}$ is the suitable X. Firstly, by the induction hypothesis, $[\![\varphi]\!] = [\![\varphi]\!]^c$, so $[\![\varphi]\!] \in \{[\![\varphi]\!]^c\}$. Secondly, for each $\psi \in \mathcal{PL}$, $[\![\psi]\!] = [\![\psi]\!]^c$ by the induction hypothesis. If $[\![\psi]\!] \in \{[\![\varphi]\!]^c\}$, it must be that $[\![\psi]\!]^c = [\![\psi]\!] = [\![\varphi]\!]^c$. By Lemma 3.1(3), it follows that $\vdash_{\mathbf{LK}} \varphi \leftrightarrow \psi$. Since $K\varphi \in w$ and $\vdash_{\mathbf{LK}} K\varphi \rightarrow B\varphi$, $B\varphi \in w$. Therefore $B\psi \in w$. So it suffices to show that $\mathcal{M}^c, w \models B\psi$, which is a standard result for a normal modal logic. Therefore, $\mathcal{M}^c, w \models K\varphi$. □

With the above lemma, a standard argument leads us to the following theorem.

THEOREM 3.4
(soundness and completeness).
For any formula φ and any set Γ of formulas, $\Gamma \vdash_{\mathbf{LK}} \varphi$ if and only if $\Gamma \models \varphi$.

4 Computational complexity

We study the computational complexity of the model checking and satisfiability problems of LK. We show that the model checking problem is in P, and the satisfiability problem is NP-complete.

4.1 The model checking problem

In this section, we consider the complexity of model checking for LK. The *model checking problem* is to determine whether a given formula is satisfied at a given state of a given model.

For a normal modal logic, typically the size of an input model is regarded as the number of states of the model. This is because even if we take the size of a binary relation into account, which is quadratic in the number of states, we typically get a similar complexity result. This is, however, not the case in our logic. Referring to [46], we define the size of our models as follows:

DEFINITION 4.1
(input size).
The size $|M|$ of a finite model $M = (W, R, S, V)$ is defined as

$$|M| = |W| + |\{(w, v) \in W \times W \mid wRv\}| + \sum_{w \in W} \sum_{X \in S(w)} \sum_{Y \in X} |Y| + |dom(V)|$$

where $dom(V)$ is the domain of the valuation V, i.e. the propositional variables that get a valuation.

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The size $|\varphi|$ of an input formula φ is the length of φ, i.e. the number of propositional variables, propositional and modal operators occurring in φ.

While at the first glance the reader may guess that the model checking problem for LK is undecidable, as the truth definition of the operator K contains a quantification over all propositional formulas, it turns out that we can efficiently decide whether a set $Y \subseteq W$ is the truth set of a propositional formula, provided that the model is finite. Thus by deciding whether Y contains the current world and all worlds accessible from current world for each $Y \in X$, which is a truth set of a $\mathcal{PL}$ formula, we can determine whether or not the argument $X \subseteq \wp(\wp(W))$ is appropriate. Given this insight, it is not hard to develop an algorithm for the model checking problem for LK. Let us first prove a proposition that is helpful for developing the algorithm.

PROPOSITION 4.2
Given a finite model $M = (W, R, S, V)$ and a set $Y \subseteq W$, $Y \in \{[\![\varphi]\!]_M \mid \varphi \in \mathcal{PL}\}$ iff there are no $w \in Y$ and $u \in W \backslash Y$ such that for every $p \in dom(V)$, $w \in V(p)$ iff $u \in V(p)$.

PROOF. From left to right. Suppose $Y = [\![\varphi]\!]$ for some $\varphi \in \mathcal{PL}$. If there exist such w and u, it follows that there is no $\mathcal{PL}$ formula that can distinguish them. But by definition, $w \models \varphi$ and $u \not\models \varphi$, which leads to a contradiction. For the converse, under the given condition, we are supposed to find a $\mathcal{PL}$ formula φ such that $Y = [\![\varphi]\!]$. For every $w \in W$ and $p \in dom(V)$, let

$$p_w = \begin{cases} p, & \text{if } w \in V(p), \\ \neg p, & \text{otherwise.} \end{cases}$$

Let $\varphi := \bigvee_{w \in Y} \bigwedge_{p \in dom(V)} p_w$. It is easy to see that $Y = [\![\varphi]\!]$. □

By the above proposition, the function ISTRUTHSET(M, Y) given below can be used to decide whether the set Y of states is the truth set of some propositional formula.

```
function isTruthSet(M, Y)
    for each x in Y do
        for each y in W do
            if x, y satisfy the same propositional variables and y ∉ Y then
                return False
            end if
        end for
    end for
    return True
end function
```

The main algorithms are presented as Algorithms 1 and 2, which are rather self-explanatory.

THEOREM 4.3
The model checking problem for LK is in P.

4.2 The satisfiability problem

In this section, we study the complexity of satisfiability problem for LK. Using the method introduced in [53], we show that this problem is NP-complete.

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Algorithm 1 Model checking: the main procedure

Input: a model $M = (W, R, S, V)$
Input: a state $w \in W$
Input: an $\mathcal{L}$ formula φ
Output 'Yes' or 'No'
  $I \leftarrow \{\}$
  $S \leftarrow sub(\varphi)$ /∗ here $sub(\varphi)$, the set of subformulas of φ, is turned into a list $\langle \varphi_1, \ldots, \varphi_n \rangle$ such that $\varphi_n = \varphi$ and for all indices i and j, if φ_i is a subformula of φ_j then $i < j$. ∗ /
  for $i \leftarrow 1$**to** S.length **do**
    if $S[i]$in dom(V) **then**
      $I[i] \leftarrow V(S[i])$
    else if $S[i] = \neg S[j](j < i)$ **then**
      $I[i] \leftarrow W \setminus I[j]$
    else if $S[i] = I[j] \wedge I[k](j, k < i)$ **then**
      $I[i] \leftarrow I[j] \cap I[k]$
    else if $S[i] = BI[j](j < i)$ **then**
      $I[i] \leftarrow \{w \in W \mid \{x \in W \mid wRx\} \subseteq S[j]\}$
    else $S[i] = KS[j](j < i)$
      $I[i] \leftarrow \{w \in W \mid MC(M, w, S[i], I[j])\}$
    end if
  end for
  if $w \in S[n]$ **then**
    return 'Yes'
  else
    return 'No'
  end if

Algorithm 2 Model checking: supporting functions for the case of $K\psi$

function $MC(M, w, K\psi, [\![\psi]\!])$
  if $w \in [\![\psi]\!]$**and**$[\![\psi]\!] \supseteq \{x \in W \mid wRx\}$ **then**
    for each X in $S(w)$ **do**
      if $[\![\psi]\!] \in X$ **and** ISGOODARGUMENT (M, X, w) **then**
        return True
    end if
end for
  return False
else
  return False
  end if
end function

function ISGOODARGUMENT(M, X, w)
  for each Y in X **do**
    if ISTRUTHSET (M, Y) **and**$Y \not\supseteq \{x \in W \mid wRx\}$ **then**
      return False
    end if
  end for
  return True
end function

Let $M = (W, R, S, V)$ be a model. We say that (the argument function S of) M is

- *austere* , if for every $w \in W$ and $X \in S(w)$, X is a singleton.
- *coherent*, if for every $w \in W$ and $Y \subseteq W$, if there exists $X \in S(w)$ such that $Y \in X$, then $\{w\} \cup \{x \in W \mid wRx\} \subseteq Y$.

LEMMA 4.4
Let $M = (W, R, S, V)$ be a austere and coherent model. For every $\varphi \in \mathcal{L}$ and $w \in W$,

$$M, w \models K\varphi \quad \text{iff} \quad \{[\![\varphi]\!]\} \in S(w).$$

PROOF. The direction from left to right follows directly from the truth definition of K and the fact that M is austere. For the other direction, if $\{[\![\varphi]\!]\} \in S(w)$, it follows that $\{w\} \cup \{x \in W \mid wRx\} \subseteq [\![\varphi]\!]$, since M is coherent. Thus $M, w \models \varphi \wedge B\varphi$. Moreover, for every $\psi \in \mathcal{PL}$ such that $[\![\psi]\!] \in \{[\![\varphi]\!]\}$,

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$[\![\psi]\!] = [\![\varphi]\!] \supseteq \{w\} \cup \{x \in W \mid wRx\}$, namely $M, w \models B\psi$ by the truth definition. Therefore, $M, w \models K\varphi$, as desired. □

In what follows, we shall make use of the standard KD45 logic of belief. In more details, we write $\mathcal{L}^B$ to denote the sublanguage of $\mathcal{L}$ without the operator K, which is a standard single-agent modal language of belief. Given a KD45 model $M = (W, R, V)$ (i.e. a Kripke model such that R is serial, transitive and Euclidean) and a state $w \in W$. The interpretation of $\mathcal{L}^B$ formulas is defined inductively as follows:

$$\begin{array}{rl} M, w \models_K p & \text{iff } w \in V(p) \\ M, w \models_K \neg\varphi & \text{iff not } M, w \models_K \varphi \\ M, w \models_K (\varphi \wedge \psi) & \text{iff } M, w \models_K \varphi \text{ and } M, w \models_K \psi \\ M, w \models_K B\varphi & \text{iff } M, v \models_K \varphi \text{ for all } v \in W \text{ such that } wRv. \end{array}$$

As before, we use $[\![\varphi]\!]_M$ to denote the truth set of φ, i.e. $\{w \in W \mid M, w \models_K \varphi\}$. Given an $\mathcal{L}$ formula φ, we write $sub(\varphi)$ for the set of all subformulas of φ.

DEFINITION 4.5
(τ-translation).
Given an $\mathcal{L}$ formula φ, the *translation for* $sub(\varphi)$ is the function $\tau : sub(\varphi) \to \mathcal{L}^B$ such that

$$\begin{array}{rcl} \tau(p) & = & p \\ \tau(\neg\psi) & = & \neg\tau(\psi) \\ \tau((\psi \wedge \chi)) & = & (\tau(\psi) \wedge \tau(\chi)) \\ \tau(B\psi) & = & B\tau(\psi) \\ \tau(K\psi) & = & p^{K\psi} \end{array}$$

where $p^{K\psi}$ is the first new propositional variable such that $p^{K\psi} \neq \tau(\chi)$ for all $\chi \in sub(\varphi) \backslash \{K\psi\}$.[5]

Intuitively, the translation τ maps each formula of the form $K\psi$ to a new distinctive propositional variable. One can verify that τ is injective.

DEFINITION 4.6
Given an $\mathcal{L}$ formula φ, we say a KD45 model $M = (W, R, V)$ *trans-satisfies* φ, if all of the following conditions hold:

1. There exists $w \in W$ such that $M, w \models_K \tau(\varphi)$;
2. For all $w \in W$, $M, w \models_K \bigwedge_{K\psi \in sub(\varphi)} \tau(K\psi) \to (\tau(\psi) \wedge B\tau(\psi))$;
3. For all $K\psi, K\chi \in sub(\varphi)$, if there is $v \in W$ such that $M, v \models_K \tau(K\psi) \wedge \neg\tau(K\chi)$, then $(\psi \wedge \neg\chi) \vee (\neg\psi \wedge \chi)$ is satisfiable (in LK).

We say φ is *trans-satisfiable* iff there is a KD45 model that trans-satisfies it.

LEMMA 4.7
Given a formula $\varphi \in \mathcal{L}$, φ is satisfiable iff φ is trans-satisfiable.

[5] We assume a unique way of selecting this propositional variable.

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PROOF. From left to right. Suppose there is a model $M = (W, R, S, V)$ and $w \in W$ such that $M, w \models \varphi$. We construct a KD45 model $M' = (W', R', V')$ such that:

- $W' = W$
- $R' = R$
- For any $p \in P$, if there is a $\psi \in sub(\varphi)$ such that $\tau(\psi) = p$, then $V'(p) = [\![\psi]\!]_M$. The valuations of other propositional variables may be arbitrary. (Note that V' is welldefined, since τ is injective.)

in the aim of showing that M' trans-satisfies φ. It suffices to show that M' satisfies the three conditions listed in Definition 4.6. But before that we state the following result, which can be shown by induction on ψ:

$$\text{For any } \psi \in sub(\varphi) \text{ and } u \in W, M, u \models \psi \text{ iff } M', u \models_K \tau(\psi). \qquad \dagger$$

With the above, we now show that the model M' satisfies the conditions 1–3.

1. It suffices to see $M', w \models_K \tau(\varphi)$, which follows from $M, w \models \varphi$ and (†).

2. Since $\models K\psi \to (\psi \wedge B\psi)$, we have $M, u \models K\psi \to \psi$ and $M, u \models K\psi \to B\psi$ for any $u \in W$ and $K\psi \in sub(\varphi)$. Suppose there is $v \in W'$ such that $M', v \not\models_K \tau(K\psi) \to (\tau(\psi) \wedge B\tau(\psi))$ for some $K\psi \in sub(\varphi)$. Then it is either $M', v \models_K \tau(K\psi) \wedge \neg\tau(\psi)$ or $M', v \models_K \tau(K\psi) \wedge \neg B\tau(\psi)$. In the former case, it follows that $M, v \models K\psi \wedge \neg\psi$ by (†), contradicting that $M, u \models K\psi \to \psi$ for any $u \in W$ and $K\psi \in sub(\varphi)$. In the latter, since $M', v \models_K \neg B\tau(\psi)$, there must be an $x \in W'$ with $vR'x$ and $M', x \models_K \neg\tau(\psi)$. Therefore, $M, x \models \neg\psi$ by (†), thus $M, v \not\models B\psi$ since $R' = R$. Because $M, v \models K\psi$ (note that $K\psi \in sub(\varphi)$ and thus (†) applies), it contradicts that $M, u \models K\psi \to B\psi$ for any $u \in W$ and $K\psi \in sub(\varphi)$. In words, $M', v \models_K \tau(K\psi) \to (\tau(\psi) \wedge B\tau(\psi))$ for any $K\psi \in sub(\varphi)$ and $v \in W'$, which leads to the second condition.

3. Suppose towards a contradiction that there is $K\psi, K\chi \in sub(\varphi)$ and $v \in W'$ such that $M', v \models_K \tau(K\psi) \wedge \neg\tau(K\chi)$, but $(\psi \wedge \neg\chi) \vee (\neg\psi \wedge \chi)$ is not satisfiable. Then $\models \psi \leftrightarrow \chi$, therefore $\models K\psi \leftrightarrow K\chi$ by the rule RE. In particular, $M, v \models K\psi \leftrightarrow K\chi$. By the truth conditions of $\leftrightarrow$ and (†), we can show that $M', v \models_K \tau(K\psi) \leftrightarrow \tau(K\chi)$, which leads to a contradiction.

From right to left. Suppose there is a KD45 model $M' = (W', R', V')$ trans-satisfying φ. Let $w' \in W'$ be the state such that $M', w' \models_K \tau(\varphi)$. Then, by the third condition, for any $K\psi, K\chi \in sub(\varphi)$ such that there is $v' \in W'$ with $M', v' \models_K \tau(K\psi) \wedge \neg\tau(K\chi)$, there is a model $M_{\psi,\chi} = (W_{\psi,\chi}, R_{\psi,\chi}, S_{\psi,\chi}, V_{\psi,\chi})$ and $w_{\psi,\chi}$ such that $M_{\psi,\chi}, w_{\psi,\chi} \models (\psi \wedge \neg\chi) \vee (\neg\psi \wedge \chi)$.

Let $M_1, \ldots, M_n$ be an enumeration of the models $M_{\psi,\chi}$ from the above clause, i.e. we take one model for each pair ψ, χ, where $M_i = (W_i, R_i, S_i, V_i)$ and let $w_1, \ldots, w_n$ be an enumeration of the worlds $w_{\psi,\chi}$ from the above clause, where $w_i \in W_i$. We can assume that $W_i \cap W_j = \emptyset$ for $i \neq j$ and $W_i \cap W' = \emptyset$ for every i.[6] Given such models $M_1, \ldots, M_n$, together with M', we are going to construct a model M and then it suffices to show that φ is satisfied at a state of M.

We first define an map $I : sub(\varphi) \to \wp(W)$ by $I(\psi) = [\![\tau(\psi)]\!]_{M'} \cup \bigcup_{i=1}^{n} [\![\psi]\!]_{M_i}$ for each $\psi \in sub(\varphi)$. We construct a model $M = (W, R, S, V)$ such that:

- $W = W' \cup \bigcup_{i=1}^{n} W_i$;
- $R = R' \cup \bigcup_{i=1}^{n} R_i$; since R' and all R_i's are serial, transitive and Euclidean—and mutually disjoint—R is also serial, transitive and Euclidean;
- For all $u \in W_i$ $(1 \leq i \leq n)$, $S(u) = \{\{Y\} \mid Y \subseteq W$ and there exists $K\psi \in sub(\varphi)$ such that $Y = I(\psi)$ and $M_i, u \models K\psi\}$;

[6] Since if there are M_i, M_j with $W_i \cap W_j \neq \emptyset$, we can easily construct an equivalent model M_j' such that $W_i \cap W_j' = \emptyset$.

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– For all $u \in W'$, $S(u) = \{\{Y\} \mid Y \subseteq W$ and there exists $K\psi \in sub(\varphi)$ such that $Y = I(\psi)$ and $M', u \models_K \tau(K\psi)\}$;
– For all $p \in P \cap sub(\varphi)$, $V(p) = I(p)$.

It then suffices to show the following results, by the last one of which we get $M, w' \models \varphi$ (since $w' \in I(\varphi)$ by definition), as we want.

(1) For any $u \in W_i$ $(1 \leq i \leq n)$ and $K\psi \in sub(\varphi)$, if $\{I(\psi)\} \in S(u)$ then $M_i, u \models K\psi$. For any $u \in W'$ and $K\psi \in sub(\varphi)$, if $\{I(\psi)\} \in S(u)$, then $M', u \models_K \tau(K\psi)$.
(2) The model $M = (W, R, S, V)$ is a austere and coherent model.
(3) For every $\psi \in sub(\varphi)$ and $w \in W$, $M, w \models \psi$ iff $w \in I(\psi)$.

Proof of (1). If $u \in W_i$ $(1 \leq i \leq n)$, since $\{I(\psi)\} \in S(u)$, there must be a $K\chi \in sub(\varphi)$ such that $I(\psi) = I(\chi)$ and $M_i, u \models K\chi$ by the definition of S. By the definition of I, it means that $[\![\chi]\!]_{M_i} = [\![\psi]\!]_{M_i}$. Therefore $M_i, u \models K\psi$ as well by the rule RE (note that RE preserves global truth, see Proposition 2.5).

If $u \in W'$, since $\{I(\psi)\} \in S(u)$, there must be a $K\chi \in sub(\varphi)$ such that $I(\psi) = I(\chi)$ and $M', u \models_K \tau(K\chi)$ by the definition of S. Suppose $M', u \not\models_K \tau(K\psi)$, then there must be a model $M_{\psi,\chi} = (W_{\psi,\chi}, R_{\psi,\chi}, S_{\psi,\chi}, V_{\psi,\chi})$ and $w_{\psi,\chi} \in W_{\psi,\chi}$ such that $M_{\psi,\chi}, w_{\psi,\chi} \models (\psi \wedge \neg\chi) \vee (\neg\psi \wedge \chi)$, which means that $I(\psi) \neq I(\chi)$, leading to a contradiction. Therefore $M', u \models_K \tau(K\psi)$.

Proof of (2). By construction, M is obviously austere. It suffices to show that M is coherent. Let $w \in W$ and $Y \subseteq W$ be arbitrary. If $w \in W_i$ $(1 \leq i \leq n)$ and $\{Y\} \in S(w)$, by the definition of S, there must be $K\psi \in sub(\varphi)$ such that $Y = I(\psi)$ and $M_i, w \models K\psi$. It follows that $[\![\psi]\!]_{M_i} \supseteq \{x \in W \mid wRx\} \cup \{w\}$ since $M_i, w \models K\psi$ and $\models K\psi \rightarrow (\psi \wedge B\psi)$. Therefore $Y = I(\psi) \supseteq [\![\psi]\!]_{M_i} \supseteq \{x \in W \mid wRx\} \cup \{w\}$. If $w \in W'$ and $\{Y\} \in S(w)$, the proof goes in a similar way.

Proof of (3). By induction on ψ. The cases of propositional variables and propositional connectives are straightforward. We only consider the cases for B and K here.

Suppose ψ is of the form $B\chi$. For any $w \in W$, if $w \in W_i$ $(1 \leq i \leq n)$, then

$M, w \models B\chi$	iff	for any $u \in W$ such that wRu, $M, u \models \chi$	(semantics)
	iff	for any $u \in W$ such that wRu, $u \in I(\chi)$	(IH)
	iff	for any $u \in W_i$ such that wR_iu, $u \in I(\chi)$	(def. of R)
	iff	for any $u \in W_i$ such that wR_iu, $u \in [\![\chi]\!]_{M_i}$	(def. of I)
	iff	$M_i, w \models B\chi$	(semantics)
	iff	$w \in I(B\chi)$	(def. of I)

If $w \in W'$, we can show the result in a similar way.

Suppose ψ is of the form $K\chi$. For any $w \in W$, if $w \in W_i$ $(1 \leq i \leq n)$,

	$M, w \models K\chi$	
iff	$\{[\![\chi]\!]_M\} \in S(w)$	(clause (2) and Lemma 4.4)
iff	$\{I(\chi)\} \in S(w)$	(IH)
iff	$M_i, w \models K\chi$	(down by (1) and up by def. of S)
iff	$w \in I(K\chi)$	(def. of I)

The case when $w \in W'$ can be shown in a similar way. □

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LEMMA 4.8
Let φ be an $\mathcal{L}$ formula. φ is trans-satisfiable iff there is a KD45 model with at most $|\alpha| + 1$ states that trans-satisfies φ, where $\alpha := \tau(\varphi) \wedge \bigwedge_{K\psi \in sub(\varphi)} \tau(K\psi) \wedge \tau(\psi)$ and $|\alpha|$ is the length of α.

PROOF. The direction from right to left is trivial. We show the other direction. Suppose a KD45 model $M = (W, R, V)$ trans-satisfies φ, and let $w \in W$ be a state such that $M, w \models_K \tau(\varphi)$. For each $B\psi \in sub(\alpha)$ such that $M, w \models_K \neg B\psi$, there must be a $v_\psi \in W$ such that wRv_ψ and $M, v_\psi \models_K \neg\psi$. By picking up exactly one v_ψ out of every such $B\psi$, we get a set X of states. We also pick up a state v such that wRv.

Let $M' = (W', R', V')$ such that:

- $W' = \{w, v\} \cup X$ (note that $|W'| \leq |\alpha| + 1$.)
- $R' = R \cap (W' \times W')$
- $V'(p) = V(p) \cap W'$ for every $p \in P$.

One can verify that M' is a KD45 model. While the transitivity and Euclidity of R' is easy to get, to show that R' is serial, it is good to observe that any states accessible from w has access to itself (guaranteed by the Euclidity of R').

For all states $u' \in W'$ and all $\psi \in sub(\alpha)$, we can show that $M', u' \models_K \psi$ iff $M, u' \models_K \psi$, by induction on ψ. In the inductive case when ψ is of the form $B\chi$.

$$\begin{array}{lcc} M, u' \models_K B\chi & \text{iff} & M, x \models_K \chi \text{ for all } x \in W \text{ such that } u'Rx \\ & \text{only if} & M, x' \models_K \chi \text{ for all } x' \in W' \text{ such that } u'R'x' \\ & \text{iff} & M', x' \models_K \chi \text{ for all } x' \in W' \text{ such that } u'R'x' \qquad \text{(IH)} \\ & \text{iff} & M', u' \models_K B\chi. \end{array}$$

For the converse direction, if $M, u' \not\models_K B\chi$, there must be an $x \in W$ such that $u'Rx$ and $M, x \models_K \neg\chi$. Since $u' \in W'$, $u' = w$ or wRu'. In either case, we have wRx (as R is transitive). So $M, w \models_K \neg B\chi$. Then by definition there exists $v_\chi \in W'$ is such that wRv_χ and $M, v_\chi \models_K \neg\chi$. It then follows that $wR'v_\chi$ (by the definition of R') and $M', v_\chi \models_K \neg\chi$ (the induction hypothesis). Since $wR'u'$ or $u' = w$, in either case, we have $u'R'v_\chi$ (as R' is Euclidean). Therefore $M', u' \not\models_K B\chi$ as desired.

Now we show that M' trans-satisfies φ. We verify the three conditions as follows:

1. $M', w \models_K \tau(\varphi)$ since $M, w \models_K \tau(\varphi)$ and $\tau(\varphi) \in sub(\alpha)$.

2. Suppose there is $v' \in W'$ and $K\psi \in sub(\varphi)$ such that $M', v' \not\models_K \tau(K\psi) \rightarrow (\tau(\psi) \wedge B\tau(\psi))$. Then, either $M', v' \models_K \tau(K\psi) \wedge \neg\tau(\psi)$ or $M', v' \models_K \tau(K\psi) \wedge \neg B\tau(\psi)$. Note that $\tau(K\psi), \tau(\psi)$ is in $sub(\alpha)$. In the former case, it follows that $M, v' \models_K \tau(K\psi) \wedge \neg\tau(\psi)$, and so $M, v' \models_K \neg(\tau(K\psi) \rightarrow \tau(\psi))$, which conflicts with the assumption that M trans-satisfies φ (second condition). In the latter case, we get a similar contradiction.

3. Suppose there exist $K\psi, K\chi \in sub(\varphi)$ and $v' \in W'$ such that $M', v' \models_K \tau(K\psi) \wedge \neg\tau(K\chi)$, with $(\psi \wedge \neg\chi) \vee (\neg\psi \wedge \chi)$ unsatisfiable. We can show $M, v' \models_K \tau(K\psi) \wedge \neg\tau(K\chi)$ since $\tau(K\psi), \tau(K\chi) \in sub(\alpha)$, which contradicts with the assumption that M trans-satisfies φ (third condition). □

With the above lemmas we get the following result.

THEOREM 4.9
The satisfiability problem for LK is NP-complete.

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PROOF. The NP-hardness follows from that of propositional logic [20, 38]. To check the satisfiability of an input formula φ in nondeterministic time of its length, we use the following procedure $Proc(\psi)$ for any given formula ψ:

"For each stage i from 1 to $d(\psi)$, for all subformulas $\psi_1$and$\psi_2$ofψ such that $d(\psi_1 \wedge \neg\psi_2) = i$, check the satisfiability of $\psi_1 \wedge \neg\psi_2$. (Note that the function d returns the modal degree of a given formula, which was defined in the proof of Lemma 3.3.)"

By Lemmas 4.7 and 4.8, it suffices to check whether $\psi_1 \wedge \neg\psi_2$ is trans-satisfied in a KD45 model within a certain size. Note also that the size of $\{\psi_1 \wedge \neg\psi_2 \mid \psi_1, \psi_2 \in sub(\psi)\}$ is quadratic in the length of ψ. So every stage can be performed in nondeterministic polynomial time:

1. Base case: $d(\psi_1 \wedge \neg\psi_2) = 0$, i.e. $\psi_1 \wedge \neg\psi_2$ is a propositional formula. It is known that the satisfiability problem of propositional logic is in NP.
2. Suppose the satisfiability problem of formulas of modal degree $\leq i$ can be decided in NP. For a formula $\psi_1 \wedge \neg\psi_2$ to check in stage $i + 1$,
   (a) Nondeterministically guess a pointed Kripke model M, w with no more than $|\beta| + 1$ states where
   $$\beta := \tau(\psi_1 \wedge \neg\psi_2) \wedge \bigwedge_{K\chi \in sub(\psi_1 \wedge \neg\psi_2)} \tau(K\chi) \wedge \tau(\chi)$$
   (Note that the length of β is polynomial in the length of $\psi_1 \wedge \neg\psi_2$. So is the number of states of M.)
   (b) We can determine whether M is serial, transitive and/or Euclidean in polynomial time. If M satisfies all these properties, we check whether:
   $$M, w \models_K \tau(\psi_1 \wedge \neg\psi_2)$$
   and whether for all states u of M
   $$M, u \models_K \bigwedge_{K\chi \in sub(\psi_1 \wedge \neg\psi_2)} \tau(K\chi) \rightarrow (\tau(\chi) \wedge B\tau(\chi))$$
   and these can be done polynomial time of the length of $\psi_1 \wedge \neg\psi_2$ (it is known that the model checking problem of normal modal logics, including KD45, can be solved in P [13]). If both hold, go to the next step.
   (c) For each $K\chi_1, K\chi_2 \in sub(\psi_1 \wedge \neg\psi_2)$, check whether:
   (i) there exists a state v with $M, v \models_K \tau(K\chi_1) \wedge \neg\tau(K\chi_2)$
   implies
   (ii) $(\chi_1 \wedge \neg\chi_2) \vee (\neg\chi_1 \wedge \chi_2)$ is satisfiable.
   If (i) implies (ii) for all such $K\chi_1$ and $K\chi_2$, then $\psi_1 \wedge \neg\psi_2$ is satisfiable. (Note that $\chi_1 \wedge \neg\chi_2, \neg\chi_1 \wedge \chi_2$ are of modal degree $\leq i$, and this step can be finished in polynomial time.)

The procedure $Proc(\psi)$ stops at the $d(\psi)$-th stage, which is in nondeterministic polynomial time as desired. The output is the set of solutions to the satisfiability problems of $\psi_1 \wedge \neg\psi_2$, for all subformulas ψ_1 and ψ_2 of φ. To get the solution to the satisfiability problem for φ, all we need is to run $Proc(\varphi \wedge (p \wedge \neg p))$. Since both φ and $(p \wedge \neg p)$ are subformulas of $\varphi \wedge (p \wedge \neg p)$, we get the solution for $\varphi \wedge \neg(p \wedge \neg p)$, which is the same as that for φ. □

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5 Variants

In this section we first introduce a multi-agent version of LK, called LK^m, and then a variant LK in which the belief operator is interpreted in the neighbourhood-like semantics to avoid the problem of logical omniscience.

5.1 Extending to multiple agents

In this section we study the extension of LK to allow multiple agents. Let A be a finite set of agents. It is natural to introduce a multi-agent language $\mathcal{L}^m$, with its syntax given by:

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \wedge \varphi) \mid B_a\varphi \mid K_a\varphi,$$

where $p \in P$ is a propositional variable and $a \in A$.

To make a difference, we shall call the multi-agent extension of LK the logic LK^m. Definitions of the semantics can also be extended naturally from the single-agent case.

An *LK^m model* (or simply, a *model*, when there is no confusion in the context) is a tuple $M = (W, R, S, V)$ such that:

- W is a non-empty set of worlds;
- $R : A \to \wp(W \times W)$ assigns every agent a serial, transitive and Euclidean relation on W; for convenience, we write R_a for $R(a)$ for any $a \in A$;
- $S : W \to \wp(\wp(\wp(W)))$ is an argumentation function; and
- $V : P \to \wp(W)$ assigns every propositional variable a set of worlds.

As usual, M, together with a state w of it, forms a *pointed model* (M, w).

Given an LK^m model $M = (W, R, S, V)$ and a world $w \in W$, that a formula α is *satisfied* in the pointed model (M, w) (notation: $M, w \models \alpha$) is defined as follows:

$$\begin{array}{lll}
M, w \models p & \text{iff} & w \in V(p) \\
M, w \models \neg\varphi & \text{iff} & \text{not } M, w \models \varphi \\
M, w \models (\varphi \wedge \psi) & \text{iff} & M, w \models \varphi \text{ and } M, w \models \psi \\
M, w \models B_a\varphi & \text{iff} & \text{for all } u \in W, \text{ if } wR_au \text{ then } M, u \models \varphi \\
M, w \models K_a\varphi & \text{iff} & M, w \models \varphi \wedge B_a\varphi \text{ and there exists } X \in S(w) \text{such that:} \\
 & & [\![\varphi]\!]_M \in X \text{and for } \psi \in \mathcal{PL} \text{ with } [\![\psi]\!]_M \in X, M, w \models B_a\psi.
\end{array}$$

Conventions are made as in Definition 2.3.

The extension goes smoothly by equipping the knowledge and belief operators, together with their semantic counterparts, for all the agents. In fact, we can carry out in the same way to reach an axiomatization for the multi-agent logic. The axiomatization **LKm** given in Figure 2 is a sound and complete axiomatization of the logic LK^m.

THEOREM 5.1
(**LKm** soundness and completeness).
For any $\mathcal{L}^m$ formula φ and any set Γ of $\mathcal{L}^m$ formulas, $\Gamma \vdash_{\mathbf{LKm}} \varphi$ if and only if $\Gamma \models \varphi$.

The above theorem can be shown in very much the same way as for **LK**.

REMARK 5.2
(global versus local argumentation functions).
In the definition of a model for the multi-agent logic, we kept the argumentation function S the same

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Axioms:	
PC	All the instances of propositional tautologies
Km	$B_a(\varphi \to \psi) \to (B_a\varphi \to B_a\psi)$
Dm	$B_a\varphi \to \neg B_a \neg\varphi$
4m	$B_a\varphi \to B_a B_a\varphi$
5m	$\neg B_a\varphi \to B_a \neg B_a\varphi$
KBm	$K_a\varphi \to B_a\varphi$
TKm	$K_a\varphi \to \varphi$
Rules:	
Nm	from φ infer $B_a\varphi$
REm	from $(\varphi \leftrightarrow \psi)$ infer $(K_a\varphi \leftrightarrow K_a\psi)$
MP	from φ and $(\varphi \to \psi)$, infer ψ

FIGURE 2 The axiomatization **LKm**

as in a model for the single-agent logic. One may argue that we also have a reason to introduce an argumentation function for each agent:

$$S : A \to W \to \wp(\wp(\wp(W))).$$

Both options have an explanation.

For what we defined above, for an agent to achieve knowledge, an argument must be *global*, in the sense that it works as well for all other agents. What this represents is more like a mathematical proof that does not easily get refused by others. Yet a *local* argumentation function allows an agent to have its own argument for something, even though others may disagree. This is perhaps more like an argument in daily life.

Our framework allows the flexibility of making a distinction between global and local argumentation, which we expect a good impact on the approaches to reasoning about different levels of knowledge.

That said, we do not introduce a logic for knowledge based on local argumentation. One of the main difficulties is to understand what higher-order knowledge means in this case. For example, a formula $K_aK_b\varphi$ requires that agent a has an argument for $K_b\varphi$, which in turn requires that agent b has an argument for φ. But do we assume there is information adopt in common by both agents, say some simple tautologies? How can these be characterized? We leave such questions for future work.

5.2 An alternative approach

The semantics we introduced for LK still has the problem of logical omniscience for the belief operator. We choose to do so mainly to keep the base logic simple and widely known. The interpretation of the knowledge operator in our framework is in fact independent of the choice of an interpretation of the belief operator. As a result, our framework extends easily if we adopt a weaker interpretation for the belief operator that does not involve the problem of logical omniscience.

Vardi [54] took Hintikka's interpretation of knowledge/belief in terms of possibility relations to be the source of the problem of logical omniscience. He introduced a 'belief structure', based on *neighbourhood models* proposed by Montague [42, 43] and Scott [47]. In a belief structure, an agent's belief is characterized not by a set of possible worlds but rather a set of propositions known by the agent.

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The logic LK is easy to be adapted to interpret the belief operator using the neighbourhood semantics. Or more generally, the interpretation of the knowledge operator in LK does not rely on that of belief, so as to be independent of the base logic of belief.

In an alternative way, a *model* can be defined as a quadruple $M = (W, N, S, V)$ such that:

- W is a non-empty set of worlds;
- $N : W \to \wp(\wp(W))$ is a neighbourhood function; constraints can be added to characterize certain properties of belief;
- $S : W \to \wp(\wp(\wp(W)))$ is an argumentation function; and
- $V : P \to \wp(W)$ assigns every propositional variable a set of worlds.

Given a model $M = (W, N, S, V)$ and a world $w \in W$, that a formula α is *satisfied* (or *true*) in the pointed model (M, w) (notation: $M, w \models \alpha$) is inductively defined as follows:

$$\begin{array}{ll}
M, w \models p & \text{iff } w \in V(p) \\
M, w \models \neg\varphi & \text{iff not } M, w \models \varphi \\
M, w \models (\varphi \wedge \psi) & \text{iff } M, w \models \varphi \text{ and } M, w \models \psi \\
M, w \models B\varphi & \text{iff } [\![\varphi]\!] \in N(w) \\
M, w \models K\varphi & \text{iff } M, w \models \varphi \wedge B\varphi \text{and there exists } X \in S(w) \text{such that:} \\
 & \quad [\![\varphi]\!]_M \in X \text{ and for all } \psi \in \mathcal{PL} \text{ with } [\![\psi]\!]_M \in X, M, w \models B\psi.
\end{array}$$

By now we hope to have convinced the reader of the independence of interpretations of the belief and knowledge operators. We will not pursue this topic further here, partly because the base logic of belief as presented above is not well studied and widely used, especially when compared with the logic KD45.

6 Related work

In this section we compare our work with existing literature. As already mentioned in the introduction, there have been a great amount of work in the area of logics of knowledge, belief and their relationships, which connects to the research on the concepts of proof, evidence and justification, and the logics of them.

Williamson's work. Of special relevance is Williamson's work on knowledge and evidence [57, 58]. In a later development [59], he introduces a logic of knowledge that is capable of solving the problem of logical omniscience, based on the idea that 'knowing p requires safety from the falsity of p and of its epistemic counterparts'. Formally, a model is a structure (W, R, V), where R consists of triples $\langle w, w^*, f\rangle$ in which w and w^* are worlds, and f is a function mapping a formula in w to a counterpart formula of it in w^*, which may endure extra constraints. Knowledge is interpreted in a way that:

$$w \models K\varphi \text{ iff for all } \langle w, w^*, f\rangle \in R, w^* \models f(\varphi).$$

By an epistemic counterpart, he refers to a belief that is alike in various epistemically relevant respects, such as how they are formed, which circumstance they are formed in, etc. Our framework, though technically quite different, is conceptually close to this type of modeling.

Justification logic. Justification logics [3, 4, 24] are a family of modal logics with the modalities replaced with justification terms, inspired by the logic of proofs [1, 2] where a justification of a formula is largely a deductive proof of it, or more precisely, a 'proof polynomial' that encodes a proof. Fitting [24] adapted Artemov's framework using a semantic solution that can be traced back to [41]. An evidence function $\mathcal{E}$ is added to a standard Kripke model, making a quadruple $(W, R, \mathcal{E}, V)$.

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The function $\mathcal{E}$ assigns an evidence set of formulas to each proof polynomial t at each world w that consists of everything t justifies at w. The interpretation looks as follows:

$$M, w \models t : \varphi \quad \text{iff} \quad \varphi \in \mathcal{E}(w, t) \text{ and for all } u, \text{ if } wRu \text{ then } M, u \models \varphi.$$

Along the line of [24], Artemov and Nogina [6, 7] introduced justification into the framework of epistemic logic. A model there is a quintuple $(W, R, R^e, \mathcal{E}, V)$ where (W, R, V) is a standard Kripke model with the binary relation R used for interpreting potential knowledge in the standard way. $\mathcal{E}$ is an evidence function as in [24], together with an extra relation R^e, the justification of a formula φ is interpreted as follows:

$$w \models t : \varphi \quad \text{iff} \quad \varphi \in \mathcal{E}(w, t) \text{ and for all } u, \text{if } wR^e u \text{ then } M, u \models \varphi.$$

This is used to model a type of explicit knowledge.

Although epistemic justifications are mainly used to characterize only the notion of knowledge, the differences in the interpretation of implicit and explicit knowledge have a technical similarity to that of belief and knowledge in this paper. This similarity is even more obvious when implicit/potential knowledge is regarded as belief.

This work diverges from the approach of justification logic, mainly in that we interpret an argument in a *semantic* way, and the latter in a *syntactic* way. More precisely, in justification logic, an argument or justification is typically a collection or sequence of formulas, while in the logic LK presented in this paper, an argument is a collection of semantic objects that are used to characterize a collection of formulas. Two different formulas can be interpreted to be the same semantic object, such as $(p \to p)$ and $(q \to q)$ both to be the collection of all possible states, and this leads to a crucial difference between the semantic and syntactic approaches. Criticism for the semantic approach from the syntactic side can be that two syntactically different arguments (i.e. different sets of formulas) may be treated to be the same semantically (i.e. the same set of semantic objects), and thus less general in expressing variants of the concept of justification, such as proof, derivation, evidence, etc. The semantic approach has its advantage as well: it better serves the purpose of formalism, especially with the concepts of satisfaction and equivalence. In particular, two formally different arguments may mean the same, and this is harder to characterize and less straightforward in the syntactic approach.

We have chosen to construct LK using the semantic approach also because we treat an argument not in the way how a statement is proved or deduced, but rather to be situations or reasons that lead to a conclusion, which is often without a clear structure.

Logics of knowledge and belief. Now we move on to the literature on logics with both knowledge and belief, in particular the relationship between them. These can be briefly classified into three categories, with approaches possibly across borders (especially between i and ii, and between i and iii):

(i) that taking both knowledge and belief as primitive and focus on the interaction properties between the two notions;
(ii) that in which either knowledge or belief is not a primitive, but a derived notion from the other; and
(iii) that in which both knowledge and belief are or can be derived notions.

In the first category, various axioms that hold for knowledge and/or belief have been discussed already in [31]. The JTB definition of knowledge ('justified true belief') has been implemented in epistemic logic in [37]. In [34], a combined system of knowledge and belief was introduced

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based on Kripke semantics (the notions of common knowledge and belief are also studied there). In their models, two accessibility relations are used to represent epistemic uncertainty and doxastic uncertainty, respectively. The logic in [34] has some interesting validities characterizing the relationship between knowledge and belief. For example, for all agents a and formulas φ, the following are valid:

$$K_a\varphi \to B_a\varphi \text{ (and also } K_a\neg\varphi \to \neg B_a\varphi)$$
$$B_a\varphi \leftrightarrow K_aB_a\varphi \text{ and } \neg B_a\varphi \leftrightarrow K_a\neg B_a\varphi$$
$$K_a\varphi \leftrightarrow B_aK_a\varphi \text{ and } \neg K_a\varphi \leftrightarrow B_a\neg K_a\varphi$$

And $B_a\varphi \to B_aK_a\varphi$ is invalid there. Some of these principles, together with others (such that $B\varphi \leftrightarrow \neg K\neg K\varphi$), were examined in [50] and followed by [30, 33]. Also based on [50], recently [8, 9] introduced a logic of knowledge and belief using a topological style of semantics. Different logics were introduced for reasoning about knowledge and belief in a combined way in [55, 56]. More sophisticated constructions are used there, which goes further away from the topic here. In [35], formulas of the forms $K\varphi$, $C^{\varphi}\psi$ and $B^{\varphi}\psi$ are taken to be primitive, which are interpreted as 'the agent knows φ', 'the agent is certain of ψ under the evidence φ' and 'the agent believes ψ under the evidence φ', respectively. Then $B\varphi$ ('φ is believed') is defined as $B^{\top}\varphi$.

In the second category, besides the well-known JTB analysis of knowledge and many work in this strand (see, e.g. [36, 51, 58]) where knowledge is defined in terms of belief, there are also approaches defining belief out of knowledge. For example, [25] defines belief in terms of knowledge and plausibility: an agent believes φ if he knows that φ is more plausible than $\neg\varphi$. [16] defines belief in a similar fashion. It is perhaps worth noting that, though from a very different motivation, their models have a flavor of neighbourhood semantics that is similar to ours (we use a neighbourhood function that selects the truth set of not mere a single formula). In [44], Moses and Shoham suggest that belief be viewed as defeasible knowledge. In particular, they come up with a definition of belief to be knowledge-relative-to-assumptions and make a connection to the notion of nonmonotonicity. The characterization of belief is also proposed informally as differences between belief and knowledge systems [45].

Approaches in the third category include defining both knowledge and belief in terms of conditional belief [10, 15, 29]. We do not elaborate this category here, since ultimately any system defining neither knowledge nor belief in terms of the other can be viewed to be of this type.

7 Conclusion

In this paper we introduced a logic of knowledge, called LK, by which we provided with an alternative implementation of the tripartite analysis of knowledge (the JTB paradigm). Our work has a conceptual link to the research on justification logic, bringing a formal notion of justification into the definition of knowledge, yet different in that justification is used in defining knowledge in terms of true belief, instead of defining explicit knowledge in terms of implicit knowledge. Our model is a combination of the standard KD45 model for belief and an argumentation function for knowledge that is of a flavor of the neighbourhood semantics. We hope this work brings the study of epistemic logic closer to the study of knowledge.

We introduced a sound and complete axiomatization for our logic and extended it to model multiple agents. In the multi-agent setting, we had two options, to model *global* or *local* arguments. We studied a logic of global arguments and leave that of local ones for future work. We explained that the base logic of belief can easily be replaced by others to avoid the problem of logical omniscience

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of the belief operator. We did not pursue with that, yet it is of interest for the future. As the reader may observe, our logic characterizes a weak notion of knowledge, without the problem of logical omniscience, but also lacking properties such as positive and negative introspection ($K\varphi \to KK\varphi$ and $\neg K\varphi \to K\neg K\varphi$) as well as the principle for the interaction between belief and knowledge (e.g. $B\varphi \to KB\varphi$). The model we propose is capable of modeling these principles by imposing further constraints (see Remark 2.6 for an example). We leave this also for future work.

We studied the computational complexity of the model checking and satisfiability problems for LK. The model checking problem can be solved in polynomial time of the input size. We do not know of the lower bound, which we will look into in the future. We showed that the satisfiability problem for LK is NP-complete—surprisingly fast as one can get.

One of the requirements of a formula to be knowledge is to have an *argument* for it. Occasionally we used other words, such as 'proof', 'justification', 'evidence', etc., in place of 'argument', though we do not in fact impose any structures of the argument—it is simply treated as a set of formulas. This is one of the reasons we call it 'abstract argument' as in the title. Our work, however, does not have a direct link to the study of formal argumentation, in particular that of abstract argumentation framework [21]. We are interested in achieving a connection to that, perhaps by means of bringing an argumentation framework into the interpretation of knowledge, so that only the accepted ones are counted as an argument.

Also interesting to look into are group notions of knowledge and belief. It is a natural task to study the extensions of our logic by incorporating common and distributed knowledge/belief. We are interested in comparing these with the logic of knowledge and belief presented in [34], and also extensions of justification logic with common knowledge [5, 17] and distributed knowledge [27, 28]. Other interesting tasks include modeling dynamics in our framework.

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